

Here is a partial result that might be useful. First, let me suppose that the signal θ is not present. Also, let me think of this as a two-person zero-sum game where player 1 chooses a transition probability $\sigma(da|s)$, player 2 chooses a transition probability $\pi(dm|s)$, and player 1's payoff is

$$\sum_{\omega} \mu_0(\omega) \left(\alpha \int_{A \times S} u(a, \omega) \sigma(da|s) G(ds|\omega) + (1 - \alpha) \int_{A \times M \times S} u(a, \omega) \sigma(da|m) \pi(dm|s) G(ds|\omega) \right).$$

Assume that $M = S$ and restrict player 2 to strategies $\pi(dm|s)$ of the form $f(m|s)\bar{G}(dm)$, where $\bar{G} := \sum_{\omega} \mu_0(\omega) G(\cdot|\omega)$. Thus, I am restricting player 2 to strategies that, for each observed s , are absolutely continuous with respect to $\bar{G}$. I'm guessing that an equilibrium of this form, if it could be shown to exist, would be an equilibrium. The remainder of this note hinges on this guess.

Thus, player 2's (restricted) pure strategy set is the convex set of jointly measurable functions $f(m|s)$ on $S \times S$. Denote this set by F . We can write 1's payoff as,

$$\sum_{\omega} \mu_0(\omega) \left(\alpha \int_{A \times S} u(a, \omega) \sigma(da|s) G(ds|\omega) + (1 - \alpha) \int_{A \times M} u(a, \omega) \left(\int_S f(m|s) G(ds|\omega) \right) \sigma(da|m) \bar{G}(dm) \right),$$

This function is continuous in σ for each $f \in F$ when we endow 1's strategy space Σ with the topology in which $\sigma_n \rightarrow \sigma$ iff $\sigma_n G(\cdot|\omega)$ weak* converges to $\sigma G(\cdot|\omega)$ for each ω (and so also $\sigma_n \bar{G}$ weak* converges $\sigma \bar{G}$).¹ Moreover Σ is compact under this topology.

Hence, since F is convex, Mertens (1986, p.238 Corollary (B)) implies that,

$$\max_{\sigma \in \Sigma} \inf_{f \in F} U(\sigma, f) = \inf_{f \in F} \max_{\sigma \in \Sigma} U(\sigma, f).$$

This is what I am able to say for now. Perhaps it will be helpful to you.
Phil.

¹Note that continuity in this topology holds even though the second integral has an integrand that is only measurable (and so perhaps discontinuous in m). Continuity nonetheless follows since the marginal of $\sigma_n \bar{G}$ on M , namely $\bar{G}$, is constant. See Balder (1988) attached.